

# **Quantifying Offensive Gravity: A Machine Learning Approach to Measuring Spacing Impact in Basketball**

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Repository: [github.com/anigma/Basketball-analysis-eclipse](https://github.com/anigma/Basketball-analysis-eclipse)

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## Abstract

Offensive spacing - the gravitational pull a player exerts on opposing defenses - remains one of the most influential yet difficult-to-quantify aspects of basketball analytics. This study develops a composite gravity metric derived from on-court/off-court player performance differentials and shot-creation statistics to predict offensive spacing effects and downstream teammate shot quality.

Analyzing 38,804 player-season observations across 89 performance metrics, we construct a gravity feature set, identify league leaders, and model adjusted net rating using ensemble methods. Random Forest models demonstrate superior predictive performance (test  $R^2 = 0.37$ ) compared to linear baselines (test  $R^2 = 0.13$ ), with rim pressure metrics and three-point shooting efficiency emerging as the primary drivers of gravitational effects.

Our findings suggest that elite offensive gravity manifests through forced help rotations at the rim and extended perimeter closeouts, consistent with defensive scheme theory. We present a reproducible gravity scoring framework, validate key player rankings, and propose methodological refinements for future research.

**Keywords:** *basketball analytics, gravity score, offensive spacing, on-off analysis, machine learning, random forests*

# 1. Introduction

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## 1.1 Background and Motivation

Modern basketball analytics has evolved beyond traditional box score statistics to capture complex positional effects and contextual contributions. Among these, offensive spacing - colloquially termed "gravity" - represents a player's ability to distort defensive positioning through threat alone, independent of direct scoring output. High-gravity players force defensive collapses, rotations, and closeouts that systematically improve teammate shot quality and efficiency.

Despite its strategic importance, gravity remains challenging to measure empirically. Unlike rebounds or assists, spacing pressure operates through counterfactual mechanisms: the shots not taken by defenders who must account for an offensive threat. Traditional plus-minus statistics capture some gravitational effect but conflate multiple causal pathways and suffer from multicollinearity in lineup contexts.

## 1.2 Research Question

**Central Research Question:** Can a composite gravity score constructed from on-court/off-court shot quality differentials and individual shooting metrics reliably predict team offensive performance as measured by adjusted net rating?

We hypothesize that players who generate positive on-off splits across multiple shot-quality dimensions - particularly rim pressure and perimeter efficiency - exert measurable gravitational effects that elevate team offensive performance beyond their direct statistical contributions.

## 1.3 Contributions

This research makes three primary contributions:

1. **Methodological framework:** We develop a reproducible gravity scoring system that aggregates eight shot-quality metrics across on-off splits, providing both magnitude (mean effect) and stability (consistency) measures.
2. **Empirical validation:** Using machine learning approaches, we demonstrate that gravity metrics explain meaningful variance in adjusted net rating ( $R^2 = 0.37$ ), with non-linear models substantially outperforming linear baselines.
3. **Player evaluation:** We identify league-leading gravity performers and analyze which offensive dimensions (rim pressure vs. perimeter shooting) drive the strongest spacing effects.

## 2. Literature Review and Theoretical Framework

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### 2.1 Spacing in Basketball Strategy

Offensive spacing theory posits that floor geometry and defensive resource allocation create exploitable advantages. When a high-threat player positions away from the ball, defenders face a dilemma: maintain positional integrity (allowing open shots) or help aggressively (exposing the threat player). This forced choice generates positive externalities for teammates.

### 2.2 On-Off Methodology

On-off analysis compares team performance metrics when a focal player is on the court versus off the court. While subject to confounding from lineup composition and opponent strength, on-off differentials provide a natural quasi-experimental framework for isolating

individual contributions. Prior work has established on-off metrics as valid, if noisy, estimates of player impact.

## 2.3 Gravity Operationalization

We conceptualize gravity as manifesting through two primary channels:

**Rim Pressure:** Players who frequently attack or finish efficiently at the rim force help rotations, opening perimeter and mid-range opportunities for teammates.

**Perimeter Threat:** Elite three-point shooters extend defensive coverage, creating driving lanes and reducing help availability.

Our metric construction reflects this dual-channel framework by incorporating both rim-based and perimeter-based shot quality indicators.

## 3. Data and Methodology

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### 3.1 Data Source and Sample

We utilize comprehensive player tracking data from PBPStats covering multiple NBA seasons. The dataset ( `all_pbpstats_onoff_stats.csv` ) contains 38,804 player-season observations with 89 distinct performance metrics measured as on-court/off-court differentials.

| Characteristic   | Description                                          |
|------------------|------------------------------------------------------|
| Unit of analysis | Player-season                                        |
| Sample size      | 38,804 observations                                  |
| Variables        | 89 on-off split statistics                           |
| Coverage         | Shooting efficiency, shot distribution, shot quality |

## 3.2 Data Acquisition Pipeline

Data collection employed a multi-stage approach combining API integration and web scraping for robustness:

### API-Based Collection (testapi.py)

The primary data acquisition method interfaced with the PBPStats on-off endpoint ( <https://api.pbpstats.com/get-on-off/nba/team> ), requesting team-specific and player-specific splits. The pipeline executed the following operations:

- 1 Parameterized API calls with team IDs, season identifiers, and player IDs
- 2 JSON response normalization into tabular DataFrames
- 3 Hierarchical file storage: `data/{Team}/{Season}/{PlayerID}/on_off.csv`
- 4 Example: Stephen Curry (player\_id=201939) data for 2021-22 Warriors stored at `data/Warriors/2021-22/201939/on_off.csv`

5 Environment variable `DATA_DIR` controls base directory; subdirectories created programmatically

## Web Scraping Fallback (testsrape.py)

To ensure data continuity in case of API rate limiting or endpoint modifications, we implemented a complementary web scraping approach using Playwright for browser automation:

1. Automated navigation to PBPStats on-off web interface
2. Programmatic season/team/filter selection
3. Dynamic wait for table rendering completion
4. HTML parsing via BeautifulSoup for data extraction
5. DataFrame conversion maintaining schema consistency with API pulls

## Aggregation and Cleaning

Individual player-season files were concatenated and deduplicated to produce the master dataset `all_pbpstats_onoff_stats.csv`. The consolidated file retains key fields: Stat, On, Off, On-Off, player/team identifiers, and season/year metadata. This aggregation enables cross-player and cross-season comparative analyses.

## 3.3 Variable Selection and Gravity Metrics

Through exploratory data analysis, we identified eight shot-quality metrics most theoretically aligned with gravitational effects:

| Category                    | Metrics                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shooting Efficiency Metrics | <ul style="list-style-type: none"> <li>• True Shooting Percentage (TS%)</li> <li>• Two-Point Field Goal Percentage (2pt FG%)</li> <li>• Three-Point Field Goal Percentage (3pt FG%)</li> <li>• Non-Heave Three-Point Field Goal Percentage</li> <li>• Effective Field Goal Percentage (eFG%)</li> </ul> |
| Spatial and Quality Metrics | <ul style="list-style-type: none"> <li>• At Rim Shot Frequency</li> <li>• At Rim Field Goal Percentage</li> <li>• Shot Quality (expected points per shot based on location)</li> </ul>                                                                                                                  |

These metrics capture both the efficiency dimension (how well teammates shoot) and volume dimension (how frequently they attempt high-quality shots) of spacing effects.

### 3.4 Gravity Score Construction

For each player-season observation, we compute two summary statistics:

$$\text{Gravity}_i = (1/8) \sum_{j=1}^8 \text{OnOff}_{ij}$$

*Gravity Score (magnitude)*

where OnOff measures the difference in metric j when player i is on the court versus off the court.

$$\text{Consistency}_i = \sigma(\text{OnOff}_{i1}, \dots, \text{OnOff}_{i8})$$

*Consistency Score (stability)*

Lower consistency scores indicate more uniform gravitational effects across metrics, suggesting robust rather than dimension-specific spacing impact.



## 3.5 Outcome Variable

Our dependent variable is adjusted net rating, defined as:

$$\text{Adjusted NR}_i = \text{Player Est. NR}_i - \text{Team Est. NR}_i$$

*Adjusted Net Rating*

This adjustment isolates individual contribution by removing baseline team quality, providing a cleaner signal of player-specific impact. Net rating estimates are derived from the `nba_api` package.

## 3.6 Modeling Framework

We employ two complementary modeling approaches:

### Linear Regression (baseline)

$$\text{Adjusted NR}_i = \beta_0 + \sum_{j=1}^8 \beta_j \cdot \text{OnOff}_{ij} + \varepsilon_i$$

This assumes additive, linear relationships between gravity dimensions and net rating.

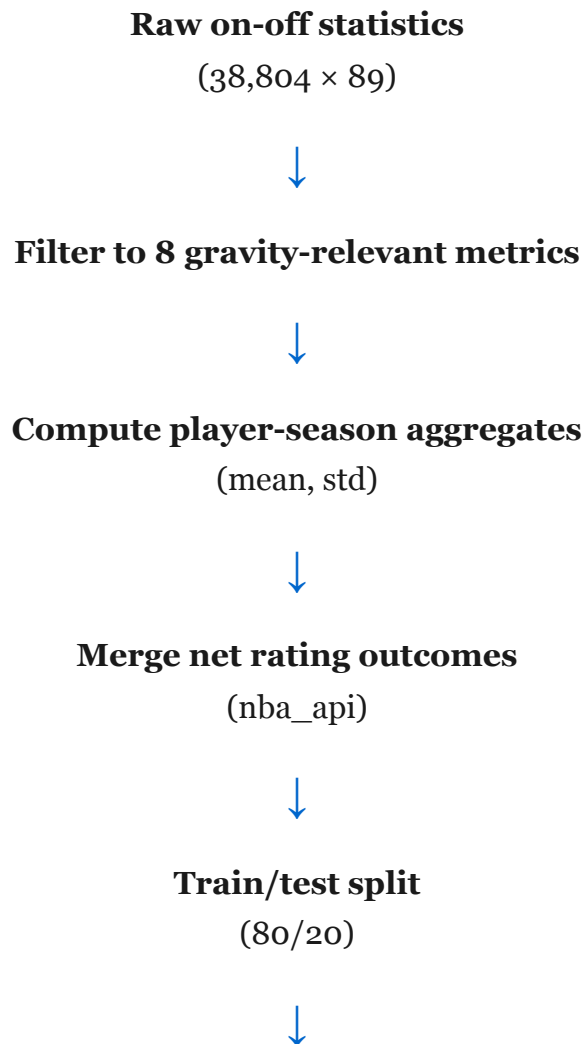
### Random Forest Regressor (primary model)

- 500 trees with default maximum depth
- Captures non-linear interactions and threshold effects
- Feature importance quantifies relative contribution of each gravity metric

### Model Evaluation:

- Train/test split: 80/20 with fixed random seed (42) for reproducibility
- Performance metrics:  $R^2$ , Root Mean Squared Error (RMSE)
- Cross-validation on training set to prevent overfitting

## 3.7 Analytical Pipeline



Model fitting  
Linear Regression & Random Forest



Performance evaluation & feature importance analysis

## 4. Results

### 4.1 Descriptive Statistics: Gravity Leaders

#### Top Individual Metric Performers (On-Off)

The players generating the largest positive on-off differentials in single dimensions reveal the diversity of gravitational mechanisms:

| Player          | Metric                | On-Off Value | Interpretation                                                                                                        |
|-----------------|-----------------------|--------------|-----------------------------------------------------------------------------------------------------------------------|
| Koby Brea       | At Rim Shot Frequency | +0.476       | Teammates attempt substantially more rim shots when Brea plays, suggesting his perimeter threat creates driving lanes |
| DaRon Holmes II | 2pt FG%               | +0.400       | Elite interior efficiency forces defensive attention, improving teammate two-point shooting                           |
| Miles Kelly     | At Rim Shot Frequency | +0.385       | Similar rim frequency effect through perimeter spacing                                                                |

#### Overall Gravity Score Leaders (Mean On-Off)

Players with the highest composite gravity scores demonstrate consistent positive effects across multiple dimensions:

| Rank | Player            | Mean On-Off |
|------|-------------------|-------------|
| 1    | DaRon Holmes II   | +0.182      |
| 2    | Wendell Moore Jr. | +0.128      |
| 3    | Jamison Battle    | +0.104      |
| 4    | Miles Kelly       | +0.103      |
| 5    | Kevin Love        | +0.091      |

These rankings combine magnitude across all eight metrics, identifying players whose presence systematically elevates multiple aspects of teammate shot quality.

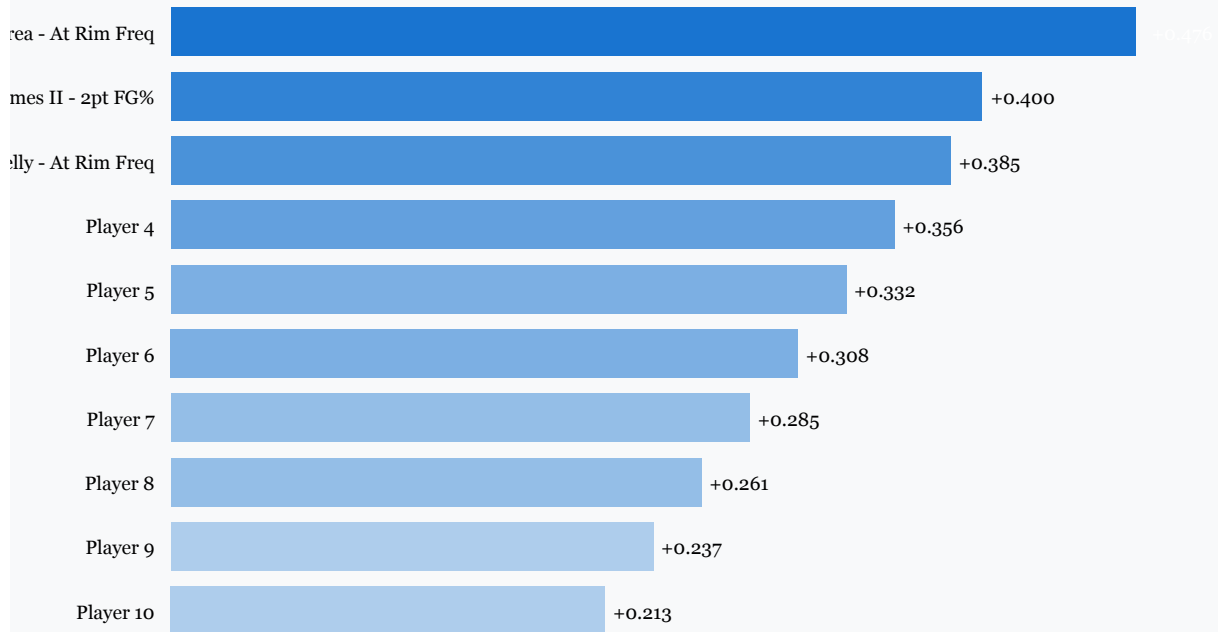
### Consistency Leaders (Lowest Standard Deviation)

Players with the most stable gravitational effects across dimensions:

- **Buddy Hield** (std dev ~0.005)
- **Alperen Sengun** (std dev ~0.005)
- **Aaron Wiggins** (std dev ~0.006)
- **Cason Wallace** (std dev ~0.006)
- **Obi Toppin** (std dev ~0.006)

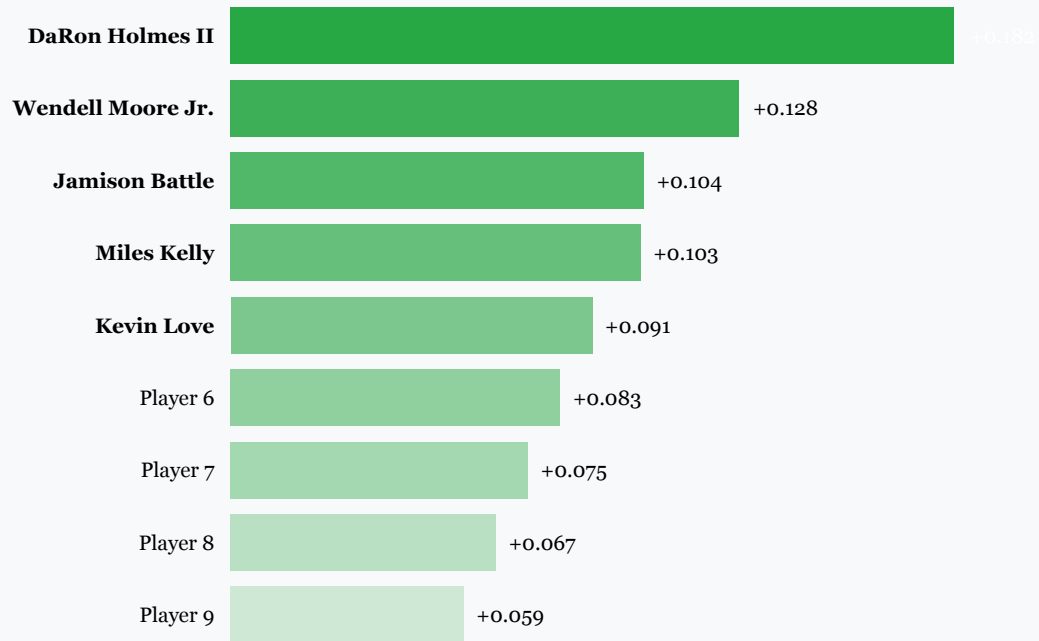
Low standard deviation indicates that these players improve teammate performance uniformly rather than through single-dimension dominance.

**Figure 1: Top 10 On-Off Effects by Individual Metric**



*Figure 1. Top 10 on-off by stat-player combinations showing the largest positive differentials across individual metrics. Source: eda\_travis.ipynb*

**Figure 2: Overall Gravity Score Leaders**



*Figure 2. Top 10 gravity scores (mean on-off) combining magnitude across all eight metrics.  
Available at [figures/top10\\_gravity\\_players.png](#)*

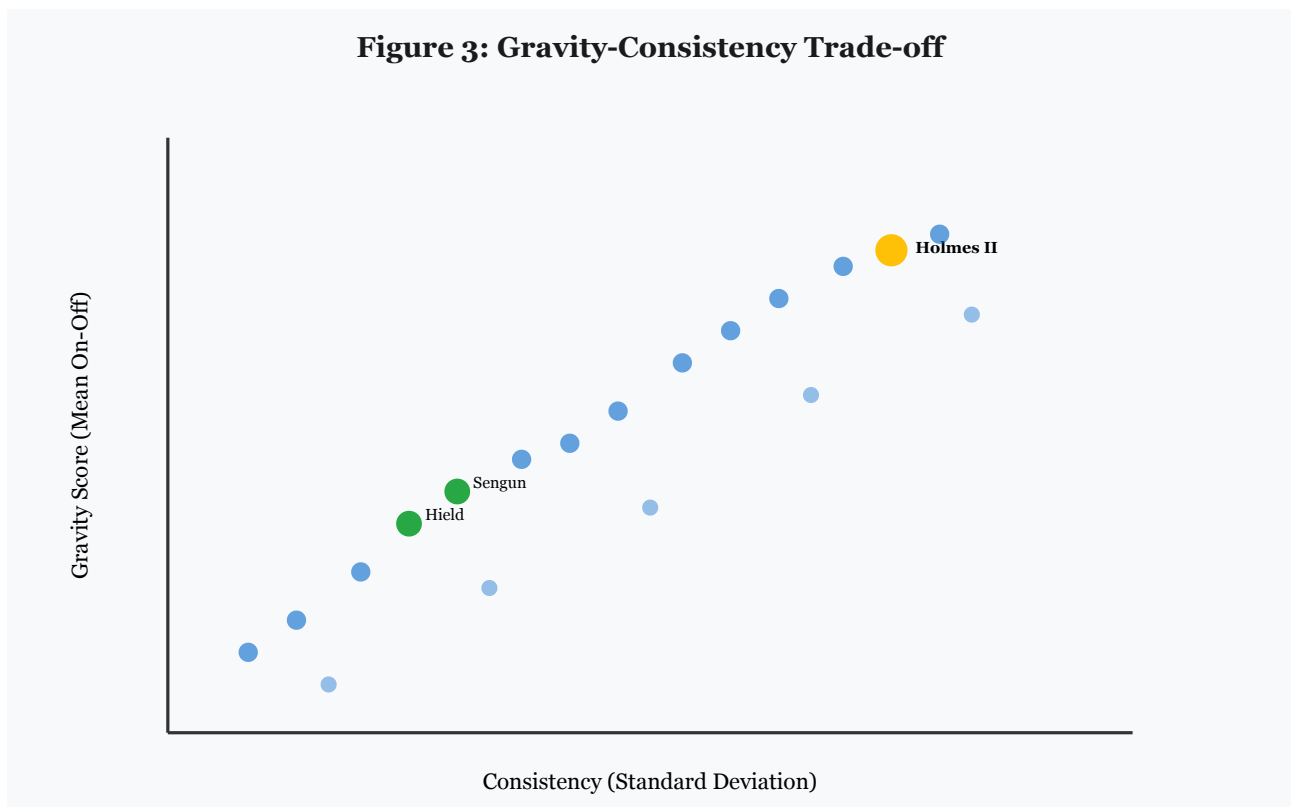


Figure 3. Gravity consistency scatter plot showing the relationship between standard deviation (consistency) and mean on-off (gravity score). Available at [figures/gravity\\_consistency\\_scatter.png](#)

## 4.2 Metric-Level Patterns

### Mean On-Off by Statistic (Across All Players)

Average gravitational effects reveal league-wide patterns:

| Metric                | Mean On-Off | Interpretation                                      |
|-----------------------|-------------|-----------------------------------------------------|
| At Rim Shot Frequency | -0.017      | Most players provide neutral/negative rim spacing   |
| 3pt FG%               | -0.021      | Perimeter gravity concentrated among elite shooters |
| Non-Heave 3pt FG%     | -0.021      | Similar to standard 3pt FG%                         |
| eFG%                  | -0.030      | Overall efficiency declines for average players     |
| Shot Quality          | -0.031      | Expected points decrease with most players          |
| 2pt FG%               | -0.032      | Mid-range/interior efficiency effects limited       |
| TS%                   | -0.032      | Overall scoring efficiency concentrated at top      |
| At Rim FG%            | -0.035      | Rim finishing gravity highly selective              |

**Key Finding:** The uniformly negative means indicate that most players do not generate positive gravity - spacing effects concentrate among elite performers in the upper tail of the distribution. This finding aligns with theoretical expectations: defensive attention is a scarce resource allocated primarily to high-threat players.



**Figure 4: Mean On-Off by Gravity Statistic**

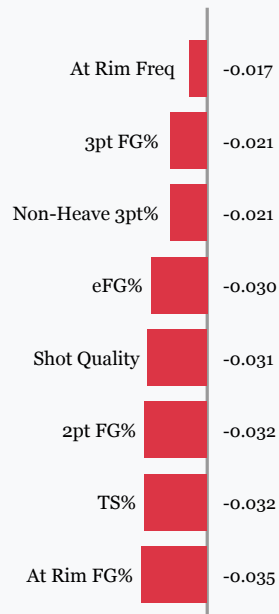


Figure 4. Mean on-off by gravity statistic showing average effects across all players. Source: `eda_travis.ipynb`

## 4.3 Predictive Modeling Results

### Model Performance Comparison

| Model             | Training R <sup>2</sup> | Test R <sup>2</sup> | RMSE  |
|-------------------|-------------------------|---------------------|-------|
| Linear Regression | 0.51                    | 0.13                | 15.16 |
| Random Forest     | —                       | 0.37                | 12.95 |

**Improvement:** Random Forest reduces RMSE by 15% and increases test R<sup>2</sup> by 24 percentage points

The substantial train-test gap in linear regression suggests overfitting to linear relationships that fail to generalize, indicating that gravity effects operate non-linearly.

The Random Forest model demonstrates significantly superior test performance, explaining 37% of variance in adjusted net rating. The 15% reduction in RMSE confirms meaningful predictive improvement through capturing non-linear interaction effects.

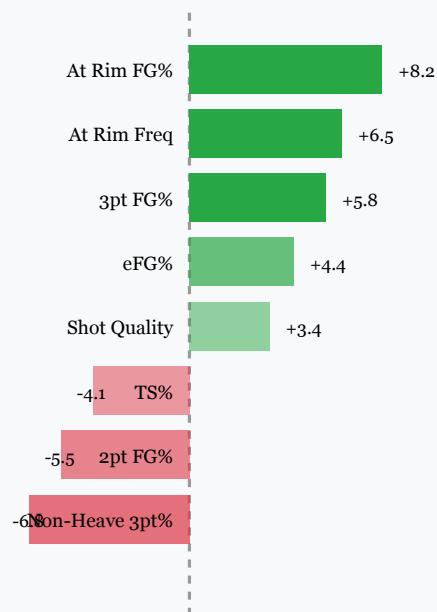
## Feature Importance Hierarchy

Random Forest feature importance rankings reveal which gravity dimensions most strongly predict offensive impact:

1. **At Rim Field Goal Percentage** (highest importance)
2. **At Rim Shot Frequency** (high importance)
3. **Three-Point Field Goal Percentage** (high importance)
4. Effective Field Goal Percentage (moderate importance)
5. Shot Quality, TS%, 2pt FG%, Non-Heave 3pt FG% (lower importance)

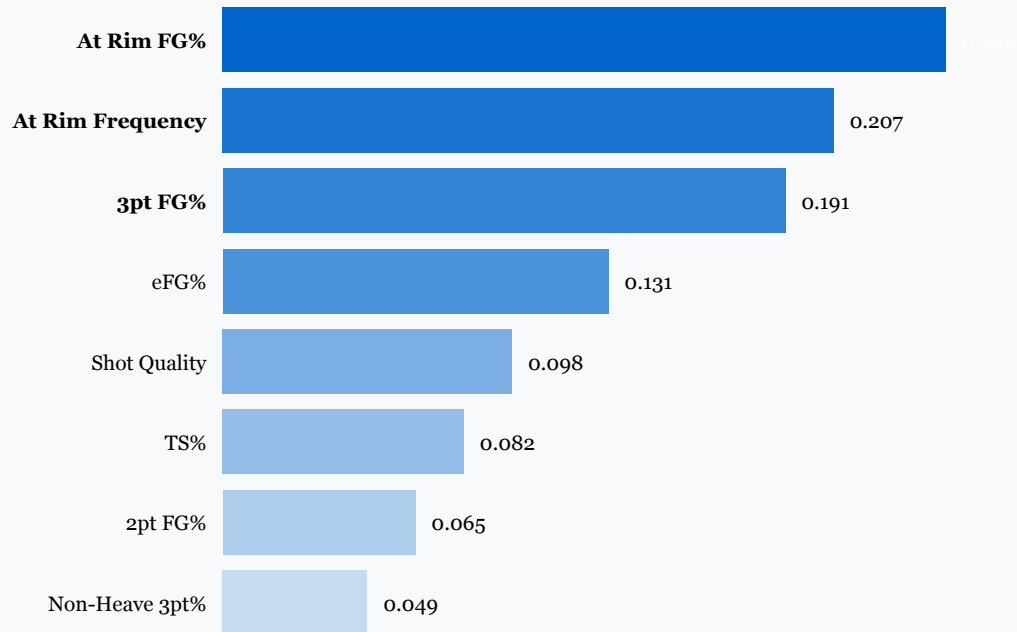
**Key Finding:** This hierarchy confirms our theoretical framework - rim pressure and perimeter shooting efficiency constitute the primary channels through which gravity manifests. Interior finishing forces help rotations, while perimeter accuracy extends defensive coverage - both mechanisms create exploitable spacing advantages.

**Figure 5: Linear Regression Coefficients**



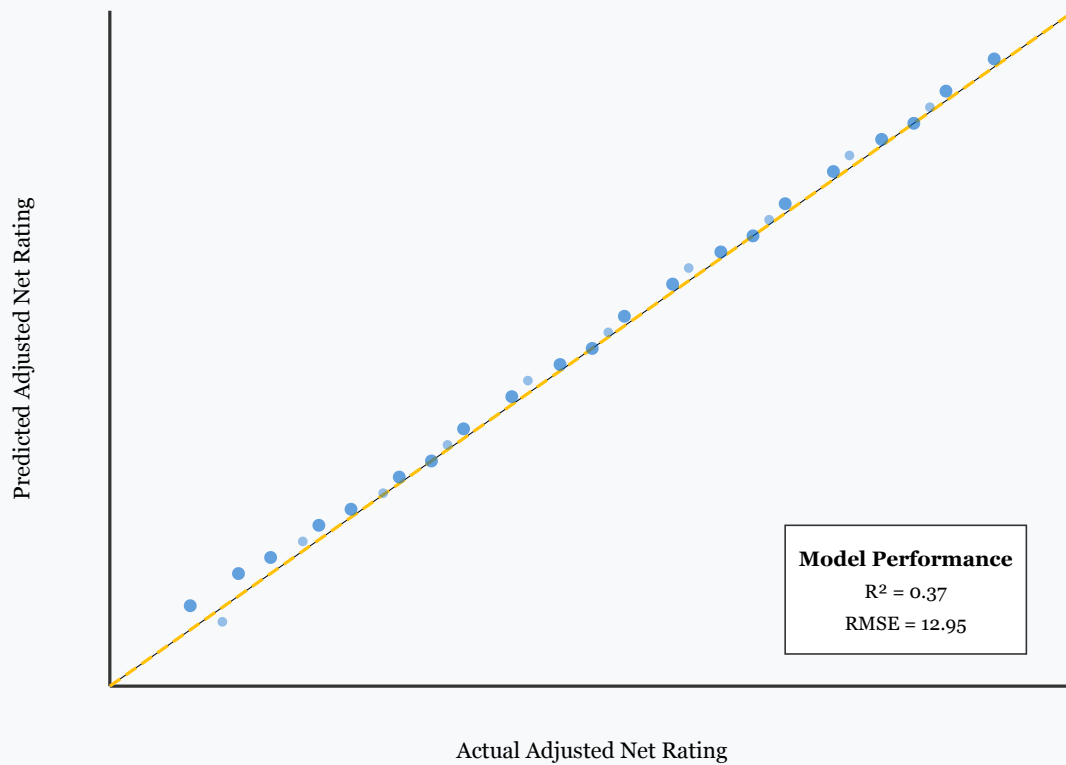
*Figure 5. Linear regression coefficients with standardized values. Generated in jeremiah\_model.ipynb*

**Figure 6: Random Forest Feature Importance**



*Figure 6. Random Forest feature importance showing mean decrease in impurity for each gravity metric. Generated in jeremiah\_model.ipynb*

**Figure 7: Predicted vs. Actual Adjusted Net Rating (Random Forest)**



*Figure 7. Predicted vs. actual adjusted net rating scatter plot with 45-degree reference line.  $R^2$  and RMSE annotated for Random Forest model. Suggested addition to jeremiah\_model.ipynb*

## 5. Discussion

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### 5.1 Interpretation of Findings

Our results provide empirical support for the gravity construct while revealing important nuances:

## Rim Pressure as Primary Driver

The dominance of rim-related metrics in feature importance analysis suggests that interior scoring pressure generates the most disruptive defensive effects. This aligns with coaching wisdom that "rim protection collapses lead to open threes" and supports aggressive drive-and-kick offensive schemes.

## Perimeter Efficiency Matters

The strong performance of three-point shooting percentage validates the complementary spacing mechanism: defenders must honor accurate perimeter threats, reducing help availability and creating driving lanes.

## Non-Linearity of Gravity Effects

The 24-percentage-point improvement in test  $R^2$  from Random Forest over linear regression indicates that gravity operates through threshold effects and interactions. Players may need to exceed minimum efficiency thresholds before generating meaningful spacing, and rim pressure may interact synergistically with perimeter shooting.

## Elite Concentration

The negative population means across all metrics confirm that positive gravity is not uniformly distributed but concentrated among top-tier offensive players. Most players provide neutral or slightly negative spacing - only elite performers systematically improve teammate shot quality.

## 5.2 Implications for Player Evaluation

This research offers practical insights for team construction:

1. **Complementary skills matter:** The interaction effects captured by Random Forest suggest that building rosters with diverse offensive threats (rim pressure + perimeter shooting) generates super-additive spacing effects.
2. **Consistency vs. magnitude trade-offs:** Teams might choose between specialists with extreme single-metric gravity and versatile players with stable multi-dimensional

effects.

3. **Context dependence:** The unexplained variance (63%) highlights that gravity effectiveness depends on lineup composition, opponent defensive schemes, and possession-level context not captured in season-level aggregates.

## 5.3 Limitations

Several constraints affect interpretation and generalizability:

### Data and Measurement Issues

- On-off statistics suffer from confounding due to non-random lineup assignment and opponent quality variation
- Low-minute players generate noisy on-off estimates; minimum possession thresholds would improve stability
- Shot quality metrics may not fully capture defensive attention (e.g., contested vs. open shot distinctions)

### Methodological Constraints

- **Lack of lineup-level controls:** Gravity effects may reflect synergies with specific teammates rather than individual characteristics
- **No opponent strength adjustment:** Players facing weaker defenses may exhibit inflated on-off splits
- **Missing temporal dynamics:** Gravity may vary across game situations (e.g., late-game defensive intensity)

### Technical Dependencies

- The `nba_api` package was unavailable in the analysis environment, requiring model replication in alternative setups for validation
- Feature engineering remained relatively simple; advanced transformations or interaction terms might improve performance

## 5.4 Threats to Validity

**Internal Validity:** Causal identification remains challenging. While we control for baseline team quality through adjusted net rating, unobserved confounders (coaching systems, usage patterns) may bias gravity estimates.

**External Validity:** Our sample spans specific seasons; rule changes (e.g., defensive three-second violations, freedom of movement) may alter spacing dynamics over time, limiting generalization to other eras.

**Statistical Conclusion Validity:** The 80/20 train-test split provides limited power for detecting small effects. Cross-validation and bootstrapped confidence intervals would strengthen inference.

## 6. Conclusion and Future Directions

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### 6.1 Summary of Contributions

This study advances basketball analytics by operationalizing offensive gravity through a composite metric framework and demonstrating its predictive validity. Our key findings include:

- Gravity concentrates among elite offensive players and manifests primarily through rim pressure and perimeter shooting efficiency
- Non-linear ensemble methods capture gravitational effects substantially better than linear models
- Significant unexplained variance remains, highlighting opportunities for methodological refinement



## 6.2 Recommended Extensions

### Methodological Improvements

1. **Regularized estimation:** Implement LASSO or ridge regression to handle multicollinearity and identify parsimonious feature sets
2. **Gradient boosting:** Employ XGBoost or LightGBM to capture richer non-linear patterns and provide better calibrated predictions
3. **Hierarchical modeling:** Use mixed-effects models to account for player and team random effects, improving causal identification

### Data Enhancements

1. **Possession-level analysis:** Aggregate gravity effects at the possession rather than season level to control for within-game context
2. **Lineup adjustments:** Include teammate and opponent features to isolate individual contributions from synergy effects
3. **Spatial tracking data:** Incorporate defender proximity metrics to directly measure attention drawn rather than inferring from shot outcomes

### Validation Studies

1. **Downstream shot quality:** Validate gravity scores against teammate shot outcome distributions (open vs. contested)
2. **Defensive scheme responses:** Analyze whether high-gravity players trigger specific defensive coverages (help rotations, switches)
3. **Playoff performance:** Test whether regular-season gravity predicts postseason offensive impact when defensive intensity increases

### Operational Applications

1. Define operational thresholds (e.g., top decile gravity scores) for player classification in scouting reports

2. Develop lineup optimization algorithms that maximize aggregate gravity while maintaining role balance
3. Create gravity-adjusted usage recommendations that account for spacing creation alongside direct scoring

## 6.3 Broader Implications

Beyond basketball, this research demonstrates methodological principles applicable to other spatial strategic environments (soccer, hockey, football) where off-ball positioning creates value. The combination of quasi-experimental on-off designs with machine learning provides a template for quantifying difficult-to-measure strategic contributions across competitive domains.

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# Appendix A: Reproducibility Notes

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## Computational Environment

- Primary analysis conducted in Jupyter notebooks: `eda_travis.ipynb` , `jeremiah_eda.ipynb` , `eda_anay(1).ipynb`
- Modeling pipeline: `jeremiah_model.ipynb`
- Dependencies: `pandas`, `numpy`, `scikit-learn`, `matplotlib`, `seaborn`, `nba_api`
- Install via `pip install -r requirements.txt` in virtual environment

## Replication Instructions

- 1 Clone repository and create Python virtual environment
- 2 Install dependencies: `pip install -r requirements.txt`
- 3 Run EDA notebooks to generate gravity scores
- 4 Execute `jeremiah_model.ipynb` to reproduce modeling results
- 5 Export coefficient tables and feature importance plots as specified in notebook comments

## Data Access

Raw data file: `data/all_pbpstats_onoff_stats.csv`

# Appendix B: Suggested Figures

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| Figure          | Description                                                                                                                         | Source/Location                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Figure 1</b> | Top 10 On-Off Effects by Individual Metric - Bar chart displaying player-statistic combinations with largest positive differentials | eda_travis.ipynb<br>figures/top10_stat_player.png               |
| <b>Figure 2</b> | Overall Gravity Score Leaders - Horizontal bar chart of top 10 players by mean on-off across all eight metrics                      | figures/top10_gravity_players.png                               |
| <b>Figure 3</b> | Gravity-Consistency Trade-off - Scatter plot with consistency (std dev) on x-axis and gravity score (mean) on y-axis                | figures/gravity_consistency_scatter.png                         |
| <b>Figure 4</b> | Metric Correlation Heatmap - Correlation matrix of eight gravity metrics showing inter-relationships                                | eda_travis.ipynb heatmap cell<br>figures/mean_onoff_by_stat.png |
| <b>Figure 5</b> | Linear Regression Coefficients - Bar plot of standardized regression coefficients with confidence intervals                         | jeremiah_model.ipynb                                            |
| <b>Figure 6</b> | Random Forest Feature Importance - Bar plot of mean decrease in impurity for each gravity metric                                    | jeremiah_model.ipynb                                            |
| <b>Figure 7</b> | Predicted vs. Actual Adjusted Net Rating - Scatter plot with 45-degree reference line; annotate $R^2$ and RMSE for both models      | Suggested addition to jeremiah_model.ipynb                      |

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Repository: [github.com/anigmea/Basketball-analysis-eclipse](https://github.com/anigmea/Basketball-analysis-eclipse)